Course Write-Up (for Courses other than External Degree Programmes)

Key items	PEIs' Inputs
Course Title: *1. If it is 100% e-learning course, add "(e-learning)" at the end of the course title.* *2. If conducted in Mandarin, add "(Mandarin)" at the end of the title.* *3. Include prefix "(MC)" for stackable certificate courses.*	**Advanced Certificate in Cybersecurity Operations Analyst**
Course Developer	√ Self-developed ☐ Externally-developed ☐ Jointly-developed *(Curriculum self-developed by Tertiary Infotech.)*
Qualification to be Awarded upon Course Completion *Note: The name of qualification to be awarded should be the same as the course title.*	**Advanced Certificate in Cybersecurity Operations Analyst**
Qualification to be Awarded by:	**Tertiary Infotech Academy**
Brief Description of Course (including learning objectives for each module)	Brief description of course and learning outcomes: The Advanced Certificate in Cybersecurity Operations Analyst equips learners with the practical, hands-on skills required to operate as front-line cybersecurity operations analysts. Upon successful completion, learners will demonstrate the ability to assess and safeguard networks, systems and applications; detect, analyse and respond to security incidents; apply principles of cybersecurity governance, risk and compliance; establish protective controls and manage vulnerabilities; and comprehend adversary behaviour in order to anticipate and counter attacks. The course comprises five modules. Each module consists of 10 sessions (9 teaching + 1 assessment), conducted live online for 3 days a week, 7:00 PM – 10:00 PM (3 hours per session). The learning objective for each module is as follows:

S/N	Module Title	Learning objective
1	**Module 1: Foundations of IT and Cloud Infrastructure**	Learners will build the technical foundation for security operations, gaining the ability to analyse computer and cloud networking, devices, ports and protocols, network segmentation and tooling; work with operating systems, databases, the command line, virtualisation/containerisation and middleware; and understand applications, APIs, automated deployment, cloud applications and scripting/coding.
2	**Module 2: Security Monitoring and Incident Handling**	Learners will be able to detect incidents using data analytics, detection use cases, indicators of compromise/attack, logs, alerts and monitoring tools; and respond to incidents through incident handling and containment, forensic analysis, malware analysis, network traffic and packet analysis, and threat analysis.
3	**Module 3: Governance, Risk and Compliance Fundamentals**	Learners will be able to apply cybersecurity principles including compliance, objectives, governance, risk management, roles and responsibilities and cybersecurity models; and assess cybersecurity risk across applications, cloud technology, data, networks, supply chain, systems/endpoints and web applications.
4	**Module 4: Protective Controls and Vulnerability Management**	Learners will be able to implement protective controls including contingency planning, identity and access management, and industry best-practice frameworks and standards; and perform vulnerability management through assessment, identification, remediation and tracking.
5	**Module 5: Threat**	Learners will be able to analyse the threat landscape

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<tr><td></td><td><table><tr><td></td><td>Intelligence and Adversary Analysis</td><td>including attack vectors, threat actors and threat intelligence sources; and understand adversary means and methods including attack types, cyber attack stages, exploit techniques and penetration testing.</td></tr></table></td></tr>
<tr><td>Target Audience</td><td>Primary Target Audience

IT and cybersecurity professionals who want to specialise in security operations and incident detection and response (e.g., SOC analysts, security engineers, system/network administrators).
Cybersecurity practitioners seeking to strengthen their threat detection, monitoring and response capabilities.
Fresh graduates from IT, computer science, data science or engineering programmes who want career-ready cybersecurity operations skills.
Mid-career professionals seeking a career switch into cybersecurity operations or blue-team roles.
GRC analysts, auditors and compliance officers who need an operational understanding of cybersecurity principles and risk.
IT support and infrastructure staff moving into security-focused roles who need to monitor, detect and respond to threats.
Technical managers or team leads who need a strong cybersecurity operations foundation to guide their teams or projects.
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<tr><td>(For certificate and above qualifications only) – Potential Employment Opportunities upon Course Completion</td><td>Graduates of the Advanced Certificate in Cybersecurity Operations Analyst can pursue a range of roles in both the public and private sectors, including:

Cybersecurity Operations Analyst
Security Operations Center (SOC) Analyst (Tier 1 / Tier 2)
Incident Detection and Response Analyst
Threat Detection / Threat Monitoring Analyst
Cyber Threat Intelligence Analyst
Incident Response Analyst
Vulnerability Management Analyst
Cybersecurity Analyst / Engineer
Security Monitoring Specialist
Cybersecurity Governance, Risk & Compliance (GRC) Analyst

These roles span industries such as finance, government, healthcare, technology and critical infrastructure, with opportunities for advancement into senior and managerial cybersecurity operations positions.</td></tr>
<tr><td>Does the proposed course curriculum follow a foreign or international curriculum through full-time programme at the Primary, Secondary, and/or High School levels (grades 1-12), including self-developed curriculum?</td><td>☐ Yes √ No
If yes, please state the location of PEI's premise for the course to be conducted in: N/A</td></tr>
<tr><td>Course Details</td><td>Mode of Delivery: ☐ Face-to-face ☐ Blended √ E-learning
(100% synchronous live virtual classes)
Duration: ~4 months (part-time)
Class Frequency: 3 days per week x 3 hours per day (part-time), 7:00 PM – 10:00 PM
Total Course Hours: 150 hours (part-time)
Computation for course hours: 5 modules x 10 sessions x 3 hours = 150 hours.
<table>
<tr><th>S/N</th><th>Course component</th><th>Number of Hours</th><th>Remarks</th></tr>
<tr><td>1</td><td>Classroom Facilitation</td><td>N/A</td><td>Course is 100% e-learning (virtual)</td></tr>
<tr><td>2</td><td>Practicum</td><td>N/A</td><td></td></tr>
<tr><td>3</td><td>Practical</td><td>69</td><td>Hands-on labs in cloud sandbox / SIEM / network & forensic analysis environments (delivered live online)</td></tr>
<tr><td>4</td><td>Synchronous e-learning</td><td>66</td><td>Live instructor-led virtual lectures & demonstrations</td></tr>
<tr><td>5</td><td>Asynchronous e-learning</td><td>N/A</td><td></td></tr>
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<tr><td>6</td><td>Assessment</td><td>15</td><td>3-hour assessment after each of the 5 modules</td></tr>
<tr><td></td><td>Total Duration (in hours)</td><td>150</td><td>50 sessions x 3 hours (45 teaching + 5 assessment)</td></tr>
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<tr><td>Minimum Entry Requirements</td><td>Age:
At least 21 years old
Academic:
At least C6 at GCE O-Level in any 3 subjects, or equivalent; or
Mature candidate who is at least 25 years old with at least 4 years of working experience</td></tr>
<tr><td>(For certificate qualifications only) – to be read in conjunction with SSG Advisory 3/2022</td><td>Will the PEI issue modular certificate(s) that students can stack towards the award of the Certificate Qualification?
√ YES ☐ NO
Note: Only “Terminal” Certificate Courses can be registered as Modular Courses that stack to a Certificate Qualification.

Maximum time allowed to complete the modular courses for award of the certificate qualification: ~4 months (part-time).
Terms and conditions to be met to be awarded the Certificate:
To be awarded the certificate, participants must:
Achieve a pass in all required assessments (one per module), and
Maintain a minimum attendance of 75% throughout the course.
<table>
<tr><th>S/N</th><th>Title of certificate qualification</th><th>Modules</th></tr>
<tr><td>1.</td><td>Advanced Certificate in Cybersecurity Operations Analyst</td><td>1. Foundations of IT and Cloud Infrastructure
2. Security Monitoring and Incident Handling
3. Governance, Risk and Compliance Fundamentals
4. Protective Controls and Vulnerability Management
5. Threat Intelligence and Adversary Analysis</td></tr>
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<tr><td>Articulation Pathway</td><td>Is the proposed course a pathway programme?
☐ YES √ NO</td></tr>
<tr><td>Information on External Course Developer
(for externally developed courses only)</td><td>Not applicable — the course is self-developed by Tertiary Infotech Academy.</td></tr>
<tr><td>Course Accreditation</td><td>Is the proposed course accredited by any external organisation?
☐ YES √ NO</td></tr>
<tr><td>Association, Collaboration or Affiliation</td><td>Is there any other form of association, collaboration or affiliation with any other organisation or persons, either local or foreign, in respect of this course?
☐ YES √ NO</td></tr>
<tr><td>Courses with Industrial Attachment</td><td>Does the proposed course have industrial attachment?
☐ YES √ NO</td></tr>
<tr><td>Other Information, if applicable</td><td>N.A.</td></tr>
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I, **Dr. ANG CHEW HOE**, (NRIC/Passport no. S1808997A) manager of **TERTIARY INFOTECH ACADEMY PTE. LTD.** declare that:

(a) the course has been approved by the Academic Board;

(b) the Academic Board has approved the deployment of teachers for the course in accordance with Regulation 26 of Private Education Regulations;

(c) the assessment methods and procedures for the course have been approved by the Examination Board; and

(d) all information provided in this application is true and accurate to my knowledge.

Alfred Ang

Date: 19 Jun 2026

(Signature of manager & date)